USB PD CTS ENGINEERING CHANGE NOTICE

NOTICE: Any Company or Companies submitting a USB Power Delivery ECR proposal must be one of the following: a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum. If a group of Companies is submitting an ECR proposal, each company must be either a Promoter or Contributor of the USB 3.0 and 2.0 Specifications who have completed the USB Power Delivery addendum.

SPECIFICATION REVISIONS AND ADDENDA: At any point in time, there shall only be one current version of the USB PD CTS, termed the production version. At the same time, there may also be proposed revisions to the specification's design which are not yet approved and shall be held confidential as deemed necessary by the USB 3.0 and USB 2.0 Promoters and within the Group of Working Committee(s).

PROCEDURES FOR SUBMITTING PROPOSALS: Both members of the USB Implementers Forum as a whole and members of the USB 3.0 and USB 2.0 Promoters may submit requests to revise the USB PD CTS Specification. Such a request may be rejected or may result in a USB PD Engineering Change Notice (ECN), which is the official way USB specifications may be changed.

FORMAT OF PROPOSAL: The originator of a request to alter the USB PD CTS Specification may do so by posting this to the USB Power Delivery Compliance working group for review. Once the proposal has been reviewed by the working group it will be passed to the USB 3.0 and 2.0 Promoters for approval to publish.

RESUBMISSION AND APPEAL: The originator of a request that was not approved can redraft the original request. Rewritten proposal will be treated as a new proposal and will be evaluated using the procedures described above. The originator of a request that was not approved can also submit an appeal to the USB 3.0 and 2.0 Promoters. The appeal must be made in writing and addressed to the Secretary of the USB Implementers Forum.

ABOUT THE ENGINEERING CHANGE REQUEST FORM:
The Purpose of this Engineering Change Request Form is to expedite the review process of the proposal by providing explanations, background information, and examples of the proposed changes at a high level. This form serves as an executive summary to the actual proposal.

STEPS ON HOW TO SUBMIT A USB PD ENGINEERING CHANGE REQUEST:

a. Please fill out the Engineering Change Request Form on the following pages completely:
    a. Detail the names and contact details for each of the ECR contributors
    b. Update the ECR Title
    c. Give a minimum of 2-3 sentences for each description on the form outlining the background to the ECR
b. For each section/table/figure to be updated:
    a. Detail the section number, starting page and figure/table number to be updated as appropriate.
    b. Detail existing text under "From Text"
    c. Detail changed text under "To Text"
c. Save the file as "USB PD CTS 1.0 R 1" followed by the ECR Title as per step a.b
d. Post the ECR in the USB PD CTS Documents section under "ECR | New ECRs".
    a. This ECR will then be reviewed by the Power Delivery Compliance Working Group.
    b. Revisions to the ECR originating from the review should be submitted as document revision of the original ECR using "Add new document".

USB PD CTS ENGINEERING CHANGE NOTICE

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USB PD CTS ENGINEERING CHANGE NOTICE

Title: COMMON.CHECK.PD.7 Check Source Capabilities Message
Applied to: USB PD CTS Specification Q3 2026

Brief description of the functional changes proposed:
- Include SPR AVS APDO in Source Capabilities check

Benefits as a result of the proposed changes:
Alignment to main specification

An assessment of the impact to the existing revision and systems that currently conform to the USB specification:
None

An analysis of the hardware implications:
none

An analysis of the software implications:
none

An analysis of the compliance testing implications:
Tester SW shall be updated

An analysis of the Vendor Info File (VIF) implications:
None

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Actual Change Requested

(a). Section x.x.x, Page x, Figure/Table x-x

To:

COMMON.CHECK.PD.7: Check Source_Capabilities Message

Description: The Tester performs additional protocol checks on all *Source_Capabilities* Messages sent by the UUT. Depending on UUT's internal power policies, the UUT may offer power that is lower than in the VIF PD_Power_As_Source field and as a result the PDOs will not match PDOs specified in the VIF. For those types of UUTs the Tester would run a reduced set of checks. On the other hand, if the UUT offers full power all the time regardless of internal or external conditions, the Tester would run the full set of checks in this test. To differentiate between two types of UUTs, the Tester uses VIF Parameter Has_Invariant_PDOs:

- If VIF Parameter Has_Invariant_PDOs is set to Y – run all checks.
- If VIF Parameter Has_Invariant_PDOs is set to N
    - and if current Source Capabilities match those in the VIF, run all checks (see note below)
    - else, if current Source Capabilities do not match those in the VIF, run all checks except those marked FULL CAPABILITY ONLY

  NOTE: There are some tests where the Tester will attempt to induce UUTs with Has_Invariant_PDOs = N to send their full capabilities. In those instances, the FULL CAPABILITY ONLY checks will be performed in accordance with the conditions above.

Check Applicability: the *Source_Capabilities* Message

Perform the following checks on all *Source_Capabilities* Messages:

1. Without receiving a SOP' *Discover Identity* Command from the UUT, the check fails if VIF field Captive_Cable = No and the UUT sends *Source_Capabilities* offering for source currents greater than 3A and/or voltages greater than 20V. [COMMON.CHECK.PD.7#1]
2. For sink, the check fails if the UUT initiates source capability message without receiving the *Get_Source_Cap* Message from the Tester. [COMMON.CHECK.PD.7#2]
3. **FULL CAPABILITY ONLY**: The Tester checks the Number of Data Objects field in the Message Header [COMMON.CHECK.PD.7#3]
    a. In PD3 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Src_PDOs.
    b. In PD2 mode, the check fails if the Number of Data Objects field in the Message Header is not equal to VIF field Num_Src_PDOs minus number of PPS PDOs and/or SPR AVS PDOs depending on what the UUT supports.
4. For the first PDO, the Tester checks the consistency [COMMON.CHECK.PD.7#4]
    a. B31…30 (Fixed Supply) set to 00b.
    b. B29 (Dual-Role Power) set to 1b if VIF field PD_Port_Type is set to Consumer/Provider, Provider/Consumer or DRP.
    c. B28 (USB Suspend Supported) set to 1b if VIF field USB_Suspend_May_Be_Cleared = NO
    d. B27 (Unconstrained Power) matches VIF field Unconstrained_Power
    e. B26 (USB Communications Capable) matches VIF field USB_Comms_Capable
    f. B25 (Dual-Role Data) set to 1b if VIF field DR_Swap_To_UFP_Supported or DR_Swap_To_DFP_Supported = YES
    g. Bit 23 (EPR Mode Capable) matches VIF field EPR_Supported_as_Src in PD3 mode.
    h. For the *Source_Capabilities* Message before the first response from the Tester
        i. Bit 24 (Unchunked Extended Messages Supported) matches VIF field Unchunked_Extended_Messages_Supported
        ii. Bits 22 (Reserved) set to 0b.
    i. For the *Source_Capabilities* Message after the Tester has sent the *Request* Message
        i. Bit 24 (Unchunked Extended Messages Supported) matches VIF field Unchunked_Extended_Messages_Supported if the test is running in PD3 mode.
        ii. Bit 22 (Reserved) set to 0b if the test is running in PD3 mode.

iii. Bits 24…22 (Reserved) set to 000b if the test is running in PD2 mode.

j. B21…20 (Peak Current) matches VIF field Src_PDO_Peak_Current1

k. B19…10 (Voltage) set to 5V.

l. B9…0 (Maximum Current)

FULL CAPABILITY ONLY: For UUT as a Source:

1. If Src_PDO_Max_current1 <= 3A, Src_PDO_Max_Current1
2. If Src_PDO_Max_current1 > 3A:
   a. The current is limited to 3A if:
      i. an emulated cable is not e-marked, or it is 3A capable only and
      ii. the Captive_Cable = NO
   b. Otherwise, the current shall be Src_PDO_Max_Current1

*NOTE: For the truth table of **Max_Current_Field**, which is* **Src_PDO_Max_current1**, see ***Error! Reference source not found.***

5. If it is not the first Fixed Supply PDO, check that B29…22 is set to zero [COMMON.CHECK.PD.7#5]
6. If it is a Programmable Power Supply APDO, check that B26…25, B16 and B7 are set to zero [COMMON.CHECK.PD.7#6]
7. If it is an Adjustable Voltage Supply APDO, check that B25…20 are set to zero [COMMON.CHECK.PD.7#13]
8. The Tester compares all PDOs to **PDP** See Error! Reference source not found. and check that they meet the requirements of the Power Rules. [COMMON.CHECK.PD.7#7]
   a. Fixed Variable PDO
      i. Check the maximum current is RoundUp (**PDP**/Voltage) or RoundDown (PDP/Voltage) to the nearest 10mA.
      ii. Check that all the normative voltages are supported by UUT based on the PDP as per the SPR Normative Voltages and Minimum Currents Table in the PD3 specification.
      iii. Check that the maximum voltage does not exceed 20V.
   b. Battery PDO
      i. Check that the maximum power is less than or equal to PDP.
   c. SPR AVS APDO
      i. Max_Current_Field:
         1. B9…0 (15V-20V) range: Check the maximum current matches 20V Fixed PDO B9…0 (Maximum Current).
         2. B19…10 (9V-15V) range: Check that the maximum current matches 15V Fixed PDO B9…0 (Maximum Current).
   d. SPR PPS APDO
      i. Check that the maximum current is RoundDown (PDP/Prog Voltage) to the nearest 50mA.
      ii. Check that all the normative SPR PPS voltages are supported by UUT based on the **PDP** as per SPR Programmable Power Supply PDOs and APDOs PDP Table in the PD3 specification if SPR PPS ADO exists.
      iii. Check that the maximum voltage does not exceed 21V.
      iv. Check that the maximum voltage and minimum voltage are as per the nominal voltages mentioned in SPR Programmable Power Supply Voltage Ranges Table in the PD3 specification.
9. **FULL CAPABILITY ONLY**: For each PDO, the Tester checks consistency of: [COMMON.CHECK.PD.7#8]
   a. Src_PDO_Supply_TypeN
   b. If Fixed Supply PDO:
      i. Src_PDO_VoltageN
      ii. Src_PDO_Peak_CurrentN
      iii. Max_current_field:

For UUT as a Source:

a. If Src_PDO_Max_currentN <= 3A, Src_PDO_Max_CurrentN

b. If Src_PDO_Max_currentN > 3A:
    i. The current is limited to 3A if:
        1. emulated cable is not e-marked, or it is 3A capable only and
        2. the Captive_Cable = NO
    ii. Otherwise, the current shall be Src_PDO_Max_CurrentN

NOTE: For the truth table of **Max_Current_Field**, which is **Src_PDO_Max_currentN**, see ***Error! Reference source not found.***

c. If Variable Supply PDO:
    i. Src_PDO_MinVoltageN
    ii. Src_PDO_MaxVoltageN
    iii. Max_current_Field:
        For UUT as a Source:
        a. If Src_PDO_Max_currentN <= 3A, Src_PDO_Max_CurrentN
        b. If Src_PDO_Max_currentN > 3A:
            i. The Current is limited to 3A if:
                1. emulated cable is not e-marked, or it is 3A capable only and
                2. the Captive_Cable = NO
            ii. Otherwise, the current shall be Src_PDO_Max_CurrentN

    NOTE: For the truth table of Max_Current_Field, which is **Src_PDO_Max_currentN**, see ***Error! Reference source not found.***

d. If Battery Supply PDO:
    i. Src_PDO_MinVoltageN
    ii. Src_PDO_MaxVoltageN
    iii. Src_PDO_Max_PowerN field:
        For UUT as a Source:
        a. If (Src_PDO_Max_PowerN / Src_PDO_MinVoltageN) <= 3A, Src_PDO_Max_PowerN
        b. If (Src_PDO_Max_Powern / Src_PDO_MinVoltageN) > 3A:
            i. The current is limited to (Src_PDO_Max_PowerN / Src_PDO_MinVoltageN) if
                1. emulated cable is not e-marked, or it is 3A capable only and
                2. the Captive_Cable = NO
            ii. Otherwise, the current shall be Src_PDO_Max_Powern / Src_PDO_MinVoltageN

    NOTE: *For the truth table of Src_PDO_Max_PowerN field*, see **Error! Reference source not found.**

e. If Adjustable Voltage Supply PDO:
    i. B27…26 (Peak Current) matches VIF field Src_PDO_Peak_Current1
    ii. Max_current_field:
        1. B9…0 (15V-20V) range: Follow the steps 9.b.iii (Fixed Supply PDO) and the Tester checks the Src_PDO_Max_CurrentN (20V Fixed PDO B9…0 , Maximum Current).
        2. B19…10(9V – 15V) range: Follow the steps 9.b.iii (Fixed Supply PDO) and the Tester checks the Src_PDO_Max_CurrentN (15V Fixed PDO B9…0, Maximum Current)

        NOTE: For the truth table of **Max_Current_Field**, which is **Src_PDO_Max_CurrentN**, see *Error! Reference source not found.* Max_Current_Field Truth Table

f. If Programmable Power Supply PDO:
    i. Src_PDO_MinVoltageN
    ii. Src_PDO_MaxVoltageN
    iii. Max_current_field:
        For UUT as a Source:

a. If Src_PDO_Max_currentN <= 3A, Src_PDO_Max_CurrentN
b. If Src_PDO_Max_currentN > 3A:
    i. The current is limited to 3A if
        1. emulated cable is not e-marked, or it is 3A capable only and
        2. the Captive_Cable = NO
    ii. Otherwise, the current shall be Src_PDO_Max_CurrentN

NOTE: For the truth table of **Max_Current_Field**, which is **Src_PDO_Max_currentN**, see ***Error! Reference source not found.***

10. The Tester checks PDOs following the first one, are in the correct order: Fixed PDOs in increasing Voltage sequence, Battery PDOs in increasing Minimum Voltage sequence, Variable PDOs in increasing Minimum Voltage sequence, then SPR AVS APDO if present and finally Programmable Power Supply APDOs if present in increasing Maximum Voltage sequence. [COMMON.CHECK.PD.7#9]
11. The Tester checks that no Fixed PDO has the same voltage as any other [COMMON.CHECK.PD.7#10]
12. The Tester checks that no Variable PDO has the same voltage range as any other [COMMON.CHECK.PD.7#11]
13. The Tester checks that no Battery PDO has the same voltage range as any other [COMMON.CHECK.PD.7#12]
14. The Tester checks that no more than one SPR AVS APDO is offered. [COMMON.CHECK.PD.7#13]